Comparative Study on the Static and Dynamic Performance of MOSFET and IGBT Devices

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I. Introduction

- Silicon carbide metal-oxide-semiconductor field-effect transistors (SiC MOSFETs) and insulated gate bipolar transistors (IGBTs) are key components in power electronics due to their high efficiency and performance.
- This study focuses on comparing the dynamic and static characteristics of these devices, analyzing dynamic characteristics like turn-on, turn-off losses, and delay times through the double pulse test and the static characteristics like breakdown voltage, transfer curve, output curve, and capacitance. Through this analysis, the study examines the switching capabilities and losses of SiC MOSFETs and IGBTs, providing valuable insights into their roles in power conversion applications.

II. Static Characteristics

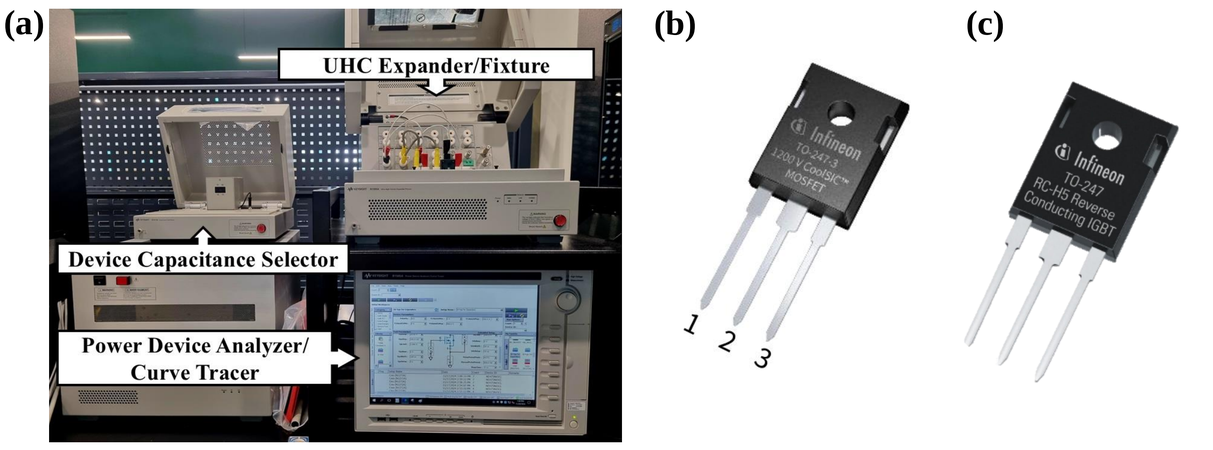


Fig. 1. (a) Static Characteristics Measurement Setup (b) 1200V SiC MOSFETs (IMW120R060M1H) (c) 1200V IGBT Device(IHW20N120R5)

▪ Static Characteristics Result

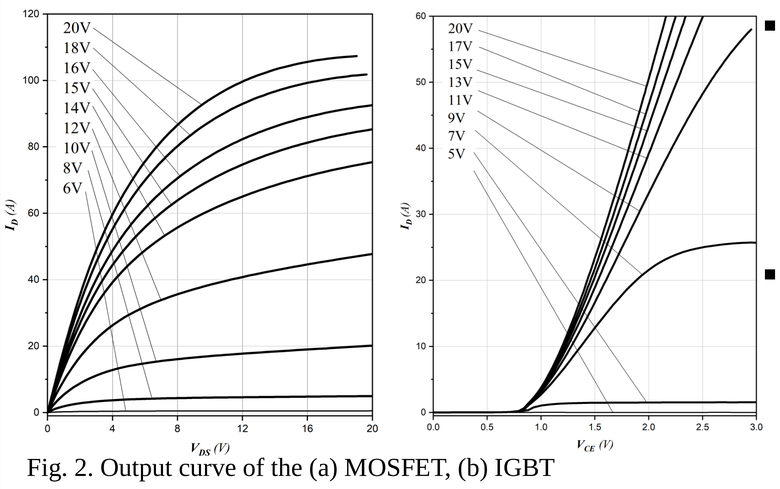


Fig. 2. Output curve of the (a) MOSFET, (b) IGBT

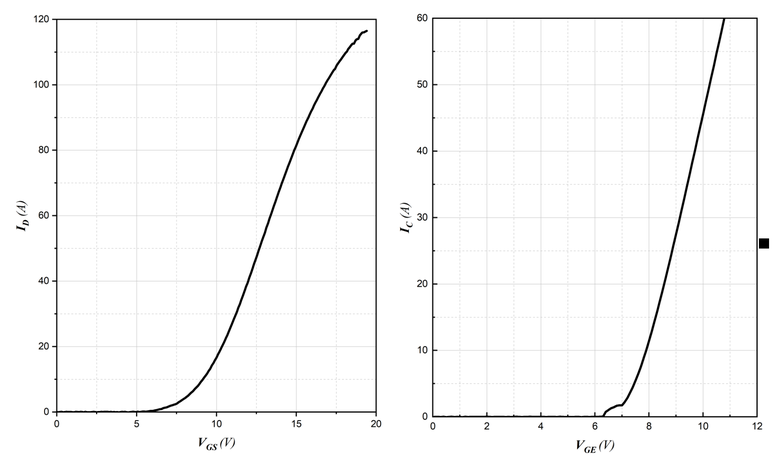


Fig. 3. Transfer curve of the (a) MOSFET, (b) IGBT

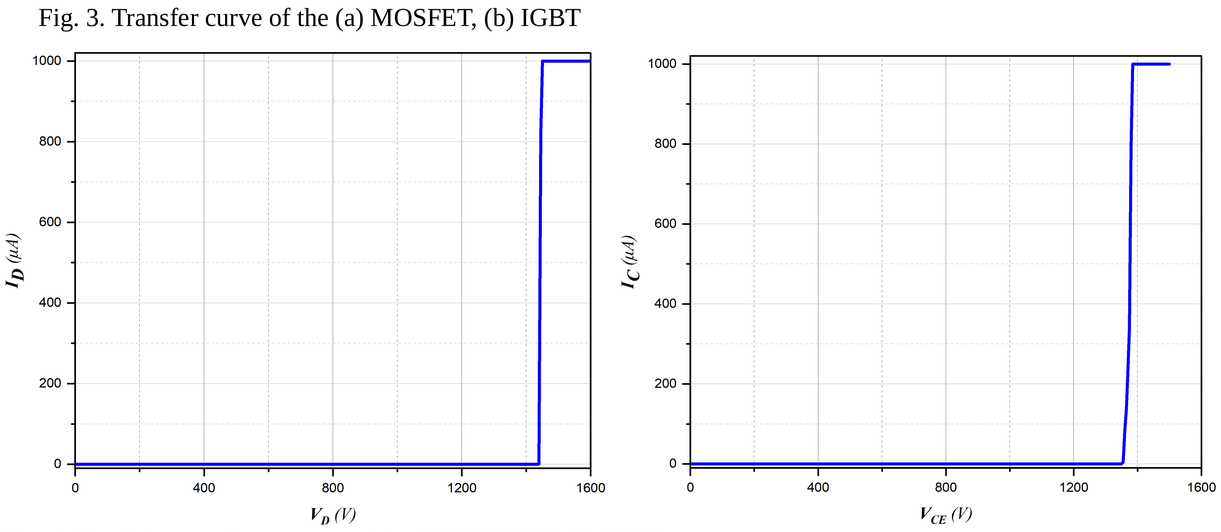


Fig. 4. Breakdown voltage curve of the (a) MOSFET, (b) IGBT

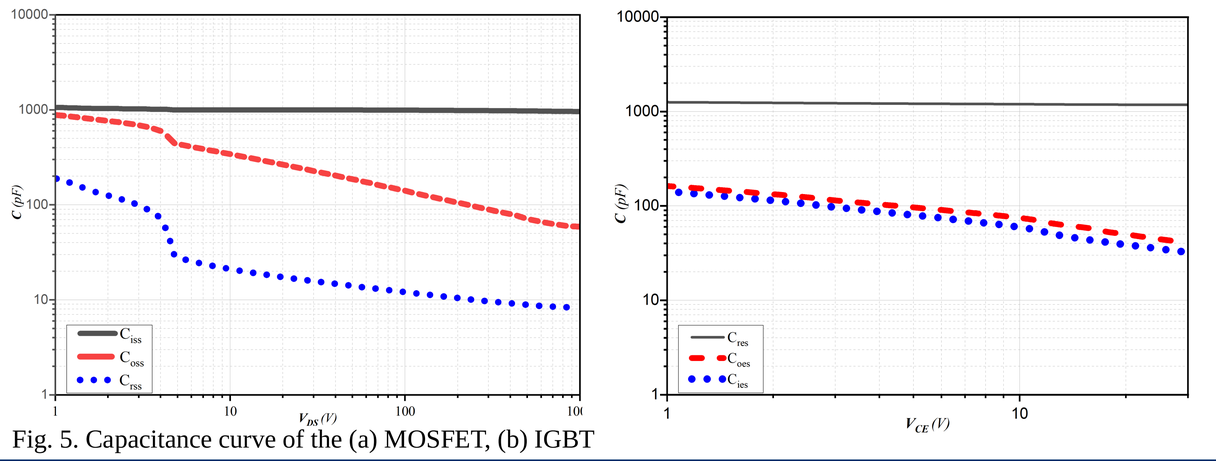


Fig. 5. Capacitance curve of the (a) MOSFET, (b) IGBT

- The static characteristics being evaluated include the output curve, transfer curve, capacitance curve, and breakdown voltage curve.
- These measurements provide insights into each devices's power handling capability, voltage tolerance, threshold voltage (V_{th}), on-resistance (R_{on}), and capacitance behavior, which are crucial for understanding its performance and reliability in steady-state conditions.
- In this analysis, we use IMW120R060M1H 1200V SiC MOSFET and IHW20N120R5 1200V IGBT from Infineon.

III. Dynamic Characteristics

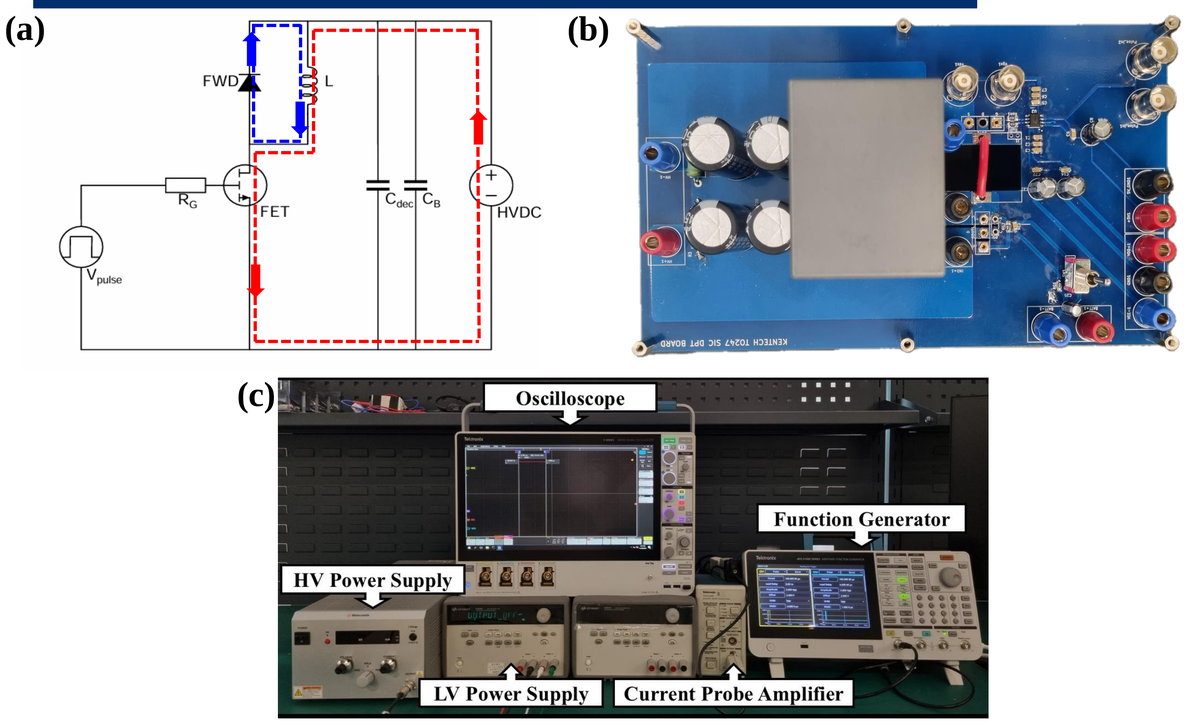
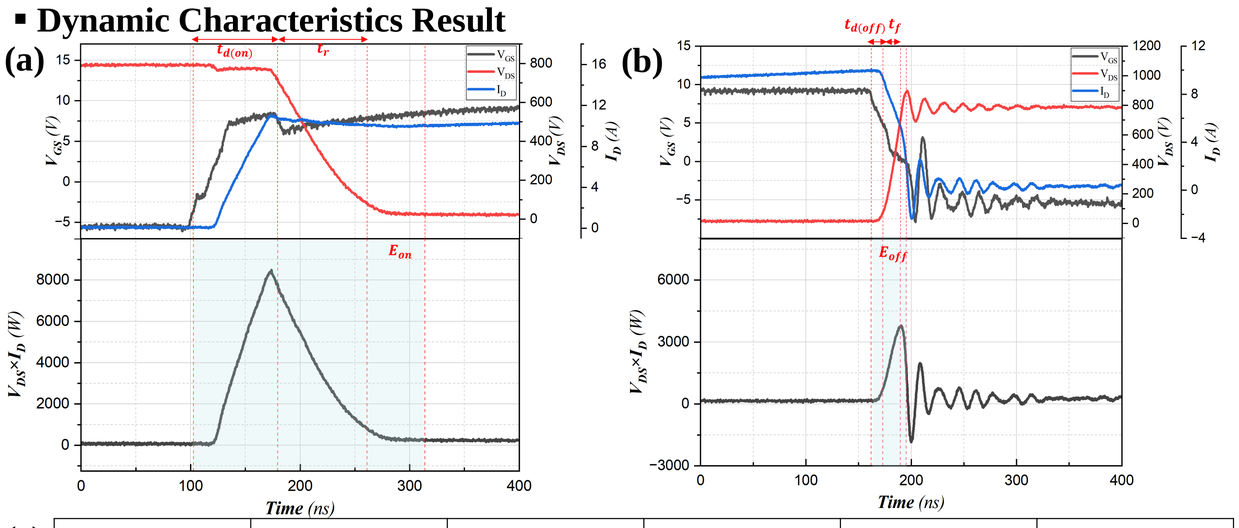


Fig. 6. (a) Double Pulse Test Circuit Diagram (b) PCB Board for Double Pulse Test (c) Double Pulse Test Setup

Device	$I_d(A)$	Period(μs)	CH1 Width(μs)	Delay(μs)	CH2 Width(μs)
MOSFET	10	200	2.750	4.750	1
IGBT	10	100	2.5	4.6	1

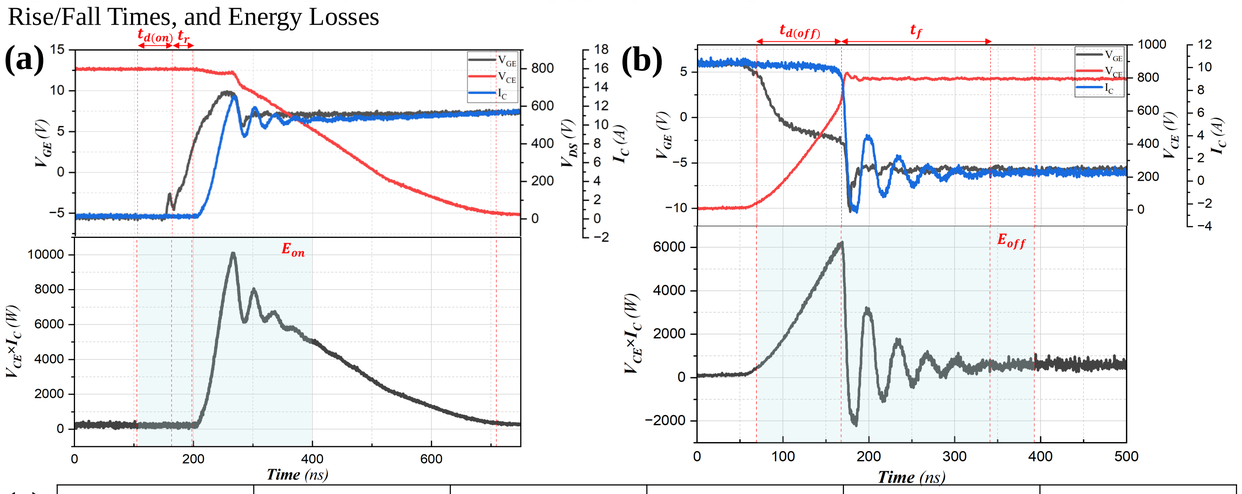
Table. 1. Parameter Values for Function Generator

▪ Dynamic Characteristics Result



(c) $t_{d(on)}(ns)$	$t_{d(off)}(ns)$	$t_r(ns)$	$t_f(ns)$	$E_{on}(\mu J)$	$E_{off}(\mu J)$
75.927	11.396	80.732	16.979	587.90	60.49

Fig. 7. (a) Turn-on Switching (b) Turn-off Switching (c) Key Switching Parameters of the MOSFET: Delay Time, Rise/Fall Times, and Energy Losses



(c) $t_{d(on)}(ns)$	$t_{d(off)}(ns)$	$t_r(ns)$	$t_f(ns)$	$E_{on}(\mu J)$	$E_{off}(\mu J)$
58.56	222.4	34.24	104.96	1838.0	363.0

Fig. 8. (a) Turn-on Switching (b) Turn-off Switching (c) Key Switching Parameters of the IGBT: Delay Time, Rise/Fall Times, and Energy Losses

- Delay time($t_{d(on)}$, $t_{d(off)}$) is the time interval between the application or removal of the gate signal and the noticeable change in current flow behavior in the device.
- Due to the parasitic inductance and capacitance on the device, delays are caused. For instance, parasitic elements such as junction-to-junction capacitance, wire inductance, and material's intrinsic resistance affect the switching characteristics.

IV. Conclusion

- The threshold of the IGBT is higher than that of the MOSFET, while the MOSFET exhibits a higher breakdown voltage than the IGBT.
- The MOSFET operates using majority carriers, whereas the IGBT utilizes both majority and minority carriers. This leads to charge storage effects and tail current in the IGBT, resulting in longer delay times as well as extended fall times during switching.
- Due to the MOSFET's faster switching characteristics compared to the IGBT, its switching losses are significantly lower than the IGBT.